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Session 1

Introduction to Number Theory

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Objectives

At the end of this session you should be able to

- Define the sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$.
- Prove the existence of irrational numbers.
- Explain the Well-Ordering Principle.
- Prove the existence or nonexistence of the least element of some subsets of real numbers.
- Prove the Archimedean Property and Mathematical Induction using the Well-Ordering Principle.
- Prove the statements on natural numbers using the Principle of Mathematical Induction.

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1.1 Introduction

Number Theory is one of the oldest branches in mathematics. Number Theory is concerned at least in its elementary aspects, with properties of the integers and more particularly with the natural numbers 1, 2, 3, ... (Also called positive integers).

Historically the origin of numbers goes back to the early age of the human civilization. A practical method of keeping accounts like, number of family members, number of cows in a flock of cows, might have kept by matching pebbles with the individuals. Later this habit might have developed as using symbols to represent each count.

We start this session with the brief introduction to the sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$, the set of Natural Numbers, the set of Integers, the set of Rational Numbers and the set of Real Numbers, respectively. After that we introduce the well ordering principle, which is play critical role in proving some elementary theorems in this session as well as the later sessions.

The Set of Natural Numbers - $\mathbb{N}$

The set of natural numbers, also called as positive integers, consist of counting numbers from 1.

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

The Set of Integers- $\mathbb{Z}$

The set of integers consists of positive integers, negative integers and the number 0.

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, 3, \dots\}$$

Observe that $\mathbb{N}$ is a proper subset of $\mathbb{Z}$. Hence $\mathbb{N} \subset \mathbb{Z}$.

The Set of Rational numbers - $\mathbb{Q}$

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The set of rational numbers can be defined as ratio of two integers with denominator is non-zero,

$$\mathbb{Q} = \left\{ \frac{p}{q} \mid p \in \mathbb{Z}, q \in \mathbb{Z} \setminus \{0\} \right\}$$

According to above definition, you can see that for a rational number we can have infinitely many equivalent representations.

For example: $\frac{1}{2} = \frac{2}{4} = \frac{8}{16} = \frac{-2}{-4} = \frac{50}{100}$ and so on.

To avoid this repetition one can also define $\mathbb{Q}$ as,

$$\mathbb{Q} = \left\{ \frac{p}{q} \mid p \in \mathbb{Z}, q \in \mathbb{N} \text{ and the greatest common factor of } p \text{ and } q \text{ is } 1 \right\}$$

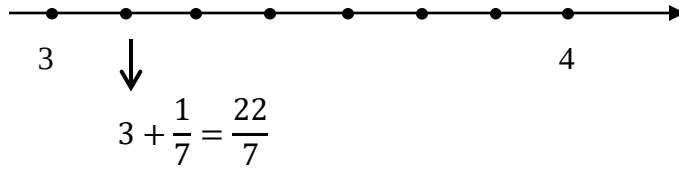
Also notice that since any $m \in \mathbb{Z}$ can be written as $\frac{m}{1}$, the set of integers is a proper subset of the set of rational numbers. Hence $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}$.

One can represent natural numbers, integers and rational numbers, in an infinitely long straight line (with no starting and ending points).

Begin with infinitely long horizontal line, locate number 0 in somewhere on it. Then by equally spacing, locate the numbers 1,2,3,4, ...and so on. As a convention, positive integers locate right to the location of point 0. To the left of the zero again, by equally spacing locates the negative integers.

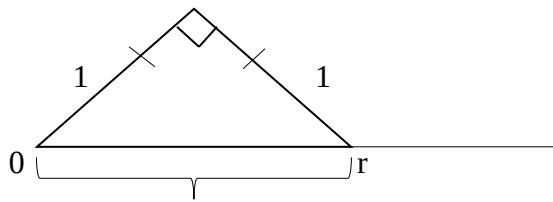
Also any rational number can be represented in the straight line. To see this, consider the already constructed points in the straight line, to represent integers. The exact middle of the points 6 and 7 corresponds to the rational number $\frac{13}{2}$. Also one can represent the number $\frac{22}{7}$ in the line, by dividing the distance between 3 and 4 into seven equal parts and from the point of 3 shifting one part to the right.

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In this manner we are convinced that for any rational number, there is a unique corresponding point in the number line.

Is it true that for a given point in the number line there is a rational number corresponds to that point? The answer is no. See the following construction.



If the hypotenuse of the isosceles right triangle is r , then that number r has the property such that $r^2 = 2$. You already know that we use the notation $\sqrt{2}$ to represent this number. If we kept the hypotenuse along the number line with one vertex of the triangle corresponds to the 0, and the other vertex right to the 0, then that point represents the number $\sqrt{2}$. But how do we know this number cannot be a rational number? In other words, how do we know that, $\sqrt{2}$ cannot be written as ratio of two integers?

The proof of this goes back to ancient Greeks. The method of proof is called contradiction method. We assume that the proposition to be proved is false and then derive a contradiction; then it follows that the original proposition must be true.

Theorem

$\sqrt{2}$ is not a rational number.

Proof:

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Assume that $\sqrt{2}$ is rational.

$$\therefore \sqrt{2} = \frac{p}{q} \text{ where } p \in \mathbb{Z}, q \in \mathbb{N}.$$

Suppose we write $\frac{p}{q}$ in its minimal representation. That is the greatest common factor of p and q is 1.

By taking the square of both sides,

$$2 = \frac{p^2}{q^2}.$$

$$\text{Hence } 2q^2 = p^2.$$

This implies that p^2 is an even number.

$\therefore p$ must be an even number. (Since if p is odd, then p^2 is an odd number.)

Therefore, let $p = 2m$, for some $m \in \mathbb{Z}$.

$$\therefore 2q^2 = (2m)^2 = 4m^2.$$

$$\therefore q^2 = 2m^2.$$

Hence q^2 is even.

$\therefore q$ is even.

$\therefore p$ is even and q is even.

This contradicts with the greatest common factor of p and q is 1.

Hence the assumption is false. Therefore $\sqrt{2}$ is not rational. ■

Note: The symbol ■ use to denote the end of a proof.

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Observing the above proof, can you prove that $\sqrt{3}$ is irrational?

The numbers which cannot be rational but can be represented on the number line is belonging to a set called irrational numbers. Hence $\sqrt{2}$ is an irrational number. After studying Sessions 3, 4 and 5, you will be realized that it is not difficult to prove that, if a positive integer is not a perfect square, then the square root of that integer is an irrational number. Hence $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\sqrt{6}$, $\sqrt{7}$, $\sqrt{8}$, $\sqrt{10}$ are examples for some irrational numbers.

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Now you know that only from the rational numbers we cannot completely fill the number line. Those “missing” points or holes called irrational numbers.

The Set of Real Numbers - $\mathbb{R}$

The set of real numbers can be defined as the union of the rational number set and the irrational number set.

According to our model (number line), any real number can be uniquely represented on the number line. Conversely, any point on the number line can be represented by a unique real number.

Finally, it is clear that $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$.

Notations: Recall the following notations you learned in PUU1141/PEU3301.

$$[a, b] = \{x | x \in \mathbb{R} \text{ and } a \leq x \leq b\}.$$

$$(a, b] = \{x | x \in \mathbb{R} \text{ and } a < x \leq b\}.$$

$$[a, b) = \{x | x \in \mathbb{R} \text{ and } a \leq x < b\}.$$

$$(a, b) = \{x | x \in \mathbb{R} \text{ and } a < x < b\}.$$

Now we are ready to discuss a very elementary but very important property about the set of natural numbers, the Well Ordering Principle. Its’ importance is enormous. We use the Well Ordering Principle to prove the several theorems in this session and as well as in subsequent sessions.

1.2 Well Ordering Principle

Before defining the Well Ordering Principle let’s define the least element of a subset of real numbers, which is useful for defining Well Ordering Principle.

1.2.1 Definition-Least Element

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Let A be a non empty subset of the set of real numbers. The real number l is called the least element of A if, $l \in A$ and for any $x \in A$, $x \geq l$.

According to the above definition you may realize that, there are some subsets of real numbers which doesn't have least elements.

1.2.2 Example:

Consider the following sub sets of the set of real numbers. Which one has a least element? Prove your answer.

- (i) $\mathbb{Z}$
- (ii) $[0, 1)$
- (iii) $(0, 1]$
- (iv) $\mathbb{N}$

Answer:

(i) It is clear that for the set $\mathbb{Z}$, it doesn't have a least element. Assume it exists. Let it be l . Since $l \in \mathbb{Z}$, $l - 1$ is also in $\mathbb{Z}$. But $l - 1 < l$, which contradict with the assumption .

(ii) The least element of $[0, 1)$ exists and is equal to 0. It is clear that $0 \in [0, 1)$ and for any $x \in [0, 1)$, $0 \leq x$.

(iii) The set $(0, 1]$ doesn't have a least element. To prove this, assume it does exist, let it be l . Then $l \in (0, 1]$. Hence $l > 0$. Now consider the real number $\frac{l}{2}$. It is clear that $0 < \frac{l}{2} < l$. Hence $\frac{l}{2}$ is smaller than l and $\frac{l}{2} \in (0, 1]$. This contradicts with the assumption.

(iv) The set $\mathbb{N}$ has a least element and is equal to 1. It is clear that $1 \in \mathbb{N}$ and for any $x \in \mathbb{N}$, $x \geq 1 \therefore 1$ is the least element of $\mathbb{N}$.

Activity. 1



Consider the following sub sets of the set of real numbers. Which one has least element? Prove your answer.

a

(i) $\{n: n \in \mathbb{N}, n > 2017\}$

(ii) $\{n: n \in \mathbb{N}, n^2 - n + 1 \geq 0\}$

(iii) $\{q: q \in \mathbb{Q}, q > 2017\}$

(iv) $\left\{\frac{1}{n}: n \in \mathbb{N}\right\}$

(v) $\{r: r \text{ is irrational and } r \geq 2\}$

(vi) $\left\{\frac{1}{n}: n \in \mathbb{Z} \setminus \{0\}\right\}$.

(vii) Find a sequence (A_n) of non empty proper subsets of $\mathbb{Z}$, each of which doesn't have a least element and satisfying $\mathbb{Z} \supset A_1 \supset A_2 \supset A_3 \supset \cdots A_n \supset A_{n+1} \supset \cdots$.

What is $\bigcap_{n=1}^{\infty} A_n$ in your sequence?

(viii) Find a sequence (A_n) of infinite proper subsets of $\mathbb{N}$, each of which satisfying

$$\mathbb{N} \supset A_1 \supset A_2 \supset A_3 \supset \cdots A_n \supset A_{n+1} \supset \cdots .$$

Does the least element exist in $\bigcap_{n=1}^{\infty} A_n$?

From above examples and self-assessment questions, you can see that there are some sub sets of real numbers which do not have least elements. But if you choose any non-empty sub set of natural numbers, then it must have a least element. This principle is called Well Ordering Principle.

1.2.3 Well-Ordering Principle

Every non-empty subset of non negative integers has a least element.

This principle plays a critical role in the proofs of this session and in subsequent sessions. Let us use it to show another property in the set of Natural Numbers known as the Archimedean Property.

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1.3 Archimedean Property

1.3.1 Theorem

For any $a, b \in \mathbb{N}$, there exists a positive integer n such that $na \geq b$.

Proof:

We use the contradiction method to prove the Archimedean Property.

Assume that the Archimedean property is not true.

Then there exist, some positive integers a and b such that $na < b$ for every $n \in \mathbb{N}$.

Let $S = \{b - na | n \in \mathbb{N}\}$.

Since $na < b$ for every $n \in \mathbb{N}$, $b - na > 0$ for every $n \in \mathbb{N}$.

But $b - na$ is an integer for any $n \in \mathbb{N}$ and hence S is a non empty subset of the natural number set.

Hence by Well Ordering Principle, S must have a least element. Let it be $b - ma$, where $m \in \mathbb{N}$.

Now since $m \in \mathbb{N}$, $m + 1 \in \mathbb{N}$. $\therefore b - (m + 1)a \in S$.

But $b - (m + 1)a = (b - ma) - a < b - ma$.

This shows $b - (m + 1)a \in S$ and $b - (m + 1)a < b - ma$, which is contradict with $b - ma$ is

The least element of S .

Hence the assumption is false. $\therefore$ The Archimedean Property is true. ■

1.3.1 Examples

(i) Let $A = \{x + y + z + v : x, y, z, v, w \in \mathbb{N}, x^2 + y^2 + z^2 + v^2 = w^3\}$. Prove that the set A has a least element.

(ii) Let $B = \{m : m \in \mathbb{N} \text{ and } m^2 \text{ is divisible by } 2, 3, 4, 5, 6, 7, 8 \text{ and } 9\}$. Prove that the set B has a least element.

[Type text]

For each of these questions, there are two ways to prove the relevant sets have least elements. One way is concretely find the least element. In this way, using some techniques, you must find The least element and need to prove that is the least element. But for these two sets it is not trivial to find the least elements. Nonetheless if you succeed to guessing the least element correctly, you need to prove that is the least element.

The other method is direct use of the Well Ordering Principle. If each of these sets is a nonempty sub sets of the natural number set, then by Well Ordering Principle least element exists. In this way, we do not concretely find the least element. We prove, merely the existence of the least element. In following answers we use the second method.

Answer:-

- (i) To apply well ordering principle we must ensure two things, the set is non empty and the set is a sub set of a natural number set. Since $x, y, z, v \in \mathbb{N}$, it is clear that $x + y + z + v$ also a member of $\mathbb{N}$. Hence the set A is a subset of $\mathbb{N}$. To prove the set A is non empty, you need some intelligent guess. Observe that $16+16+16+6 = 64$, this can be read as $4^2+4^2+4^2+4^2 = 4^3$. Then by taking $x = 4, y = 4, z = 4, v = 4$ and $w = 4$, it is clear that $x + y + z + v = 16$ is an element of A . Hence A is a nonempty subset of $\mathbb{N}$. Therefore by Well Ordering Principle the set A has a least element.
- (ii) It is trivial that B is a sub set of $\mathbb{N}$. Again, if we choose $m = 2.3.4.5.6.7.8.9$, then clearly m^2 is divisible by 2, 3, 4, 5, 6, 7, 8 and 9. Therefore 2.3.4.5.6.7.8.9 is an element of B and hence B is nonempty. Therefore by Well Ordering Principle the set B has a least element.

1.3.2 Example

Find whether the least element exist for each of the following sets. Justify your answer.

- (i) $A = \{x + y : x, y \in \mathbb{N}, 13x - 7y = 1\}$,
(ii) $B = \{x - y : x, y \in \mathbb{N}, 13x - 7y = 1\}$,
(iii) $C = \{x + y : x, y \in \mathbb{N}, 9x - 6y = 5\}$

[Type text]

Answer:-

(i)

Clearly, since $x, y \in \mathbb{N}$, $x + y \in \mathbb{N}$. Hence A is a sub set of $\mathbb{N}$. Observe that

$13 \times 6 - 7 \times 11 = 1$. Hence $6 + 11 = 17$ is a member of the set A . This shows that, the set A is a non empty subset of $\mathbb{N}$. Therefore by Well Ordering Principle the set A has a least element.

(ii)

It is clear that, some negative integers are members of B . $\{13 \times 6 - 7 \times 11 = 1$. Hence $6 - 11 = -5$ is a member of the set $B\}$. We know that, for some sub sets of $\mathbb{Z}$ have least elements and for some none empty sub sets of $\mathbb{Z}$ doesn't have.

Assume that the set B has a least element. Let it be L . Hence there exist $a, b \in \mathbb{N}$ such that $L = a - b$ and $13a - 7b = 1$. Now let $x = a + 7$ and $y = b + 13$. Then clearly $x, y \in \mathbb{N}$ and $13x - 7y = 13(a + 7) - 7(b + 13) = 13a - 7b = 1$. Hence $x - y \in B$. Also observe that $x - y = (a + 7) - (b + 13) = a - b - 6 = L - 6 < L$. Which contradict with L is the east element of B . Hence our assumption is wrong and therefore the set B doesn't have a least element.

(iii)

Again it is clear that the set C is a sub set of $\mathbb{N}$. If we can prove C is non empty, then as in (i), we can apply the Well Ordering Principle to prove the existence of the least element.

But observe that, if there are two positive integers x and y such that $9x - 6y = 5$ then

$3(3x - 2y) = 5$. Hence $(3x - 2y) = \frac{5}{3}$ this cannot happen, since $3x - 2y$ is an integer.

Hence the set C is empty. Hence C doesn't have a least element.

Activity. 2



(a) Does the set $\{113n - 355 : n \in \mathbb{N}\}$ has a least element? Justify your answer.

(b) Does the set $\{355 - 113n : n \in \mathbb{N}\}$ has a least element? Justify your answer.

(c) Guess the least element of the set $\{x + y + z : x, y, z \in \mathbb{N}, x^2 + y^2 = z^2\}$. Prove your guessing is correct.

[Type text]

- (d) According to the Well Ordering Principle, does the least element of the set $\{x : x \in \mathbb{N}, x^2 + 1 \text{ is a multiple of } 3\}$ exist? Justify your answer.

Activity. 3



Consider the following sets and complete the activity given below.

$$A = \{x + y - z : x, y, z \in \mathbb{N}, x^2 + y^2 = z^2\},$$

$$B = \{z - x - y : x, y, z \in \mathbb{N}, x^2 + y^2 = z^2\},$$

$$C = \{z + x + y : x, y, z \in \mathbb{N} \text{ and } x, y, z \text{ are odd numbers}, x^2 + y^2 = z^2\}$$

- (i) Show that $A \neq \emptyset$.
- (ii) Using a well known property in a triangle, show that any element of A is positive.
- (iii) Using algebraic methods, show that any element of A is positive.
- (iv) Show that $B \neq \emptyset$.
- (v) Prove that B doesn't have a least element. {Hint: if $5 - 4 - 3 \in B$ then does the element $5k - 4k - 3k \in B$?

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Does the set $\{x + y : x, y \in \mathbb{N}, x, y \text{ both odd and } x^2 + y^2 \text{ is a perfect square}\}$ has a least element?

1.4 Principle of Mathematical Induction

Now let us use the well ordering principle to prove the Principle of Mathematical Induction, which is familiar to you from your A/L s. Also you learned about Principle of Mathematical Induction in the course PEU3300/PUU1140. We strongly recommend you to study again the section 22.3 in Session 22 of unit II of PEU 3300/PUU1140.

In the course PEU 3300/PUU1140 you learned the theorem of Principle of Mathematical

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Induction without it's proof. Here we prove the Principle of Mathematical Induction using the Well Ordering Principle. Notice that for the convenience of applying the Well Ordering Principle, the appearance of the following theorem is slightly different from the familiar form.

1.4.1 Theorem - Principle of Mathematical Induction

Let S be a subset of positive integers (i.e. $S \subseteq \mathbb{N}$) with the following properties:

- (i) The integer 1 belongs to S . (i.e. $1 \in S$)
- (ii) Whenever the integer k in S , the next integer $k + 1$ must also be in S .

Then the set S is equal to the set of natural numbers.

Proof:

Let $T = \mathbb{N} \setminus S$. i.e. the set T consists of all positive integers which are not in S .

(Recall that if A and B are two sets then the set $A \setminus B = \{x | x \in A, x \notin B\}$.)

Assume that $T \neq \phi$. $\therefore T$ is a non empty sub set of natural numbers.

Hence by well ordering principle T has a least element. Let it be a .

From property (i), $1 \in S$.

$\therefore 1 \notin T$.

$\therefore a \neq 1$.

$\therefore a > 1$.

Now since a is the least element of T , $a - 1 \notin T$.

$\therefore a - 1 \in S$.

Now by property (ii), $(a - 1) + 1 = a \in S$. This contradicts with the fact $a \in T$.

Hence the assumption is false.

$\therefore T = \phi$

$\therefore S = \mathbb{N}$. ■

1.4.2 Example:

Use the Principle of Mathematical Induction to show that, for each $n \in \mathbb{N}$,

[Type text]

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2} \dots\dots\dots(1)$$

Proof:

Let $S = \left\{ n \mid n \in \mathbb{N}, 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2} \right\}$. i.e. S be the set of positive integers which satisfies the equation (1).

Observe that $1 = \frac{1(1+1)}{2}$ i.e. the equation (1) is true for $n = 1$.

$$\therefore 1 \in S. \dots\dots\dots(\alpha)$$

Let $k \in S$ (Where k is fixed but unspecified integer).

$$\text{Then } 1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}.$$

If we add $k + 1$ to the both sides, then $1 + 2 + 3 + \cdots + k + (k + 1) = \frac{k(k+1)}{2} + (k + 1)$.

$$\text{But } \frac{k(k+1)}{2} + (k + 1) = \frac{(k+1)(k+2)}{2} = \frac{(k+1)((k+1)+1)}{2}.$$

Hence for $n = k + 1$ the equation (1) is true. This implies $(k + 1) \in S$.

$$\therefore \text{Whenever } k \in S, \text{ the integer } k + 1 \in S. \dots\dots\dots(\beta)$$

By $(\alpha), (\beta)$ the set S satisfies the two properties of the Principle of Mathematical Induction.

$$\therefore S = \mathbb{N}.$$

$$\text{Hence for each } n \in \mathbb{N}, 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}. \blacksquare$$

Remark

When giving induction proofs, the most common method is shorten the argument by eliminating all reference to the set S , and proceed to show that the result in the problem(or its modification) is true for the integer 1, and if it is true for the integer k , then it is also true for $k + 1$.

Activity. 3



Re read the section 22.3 in Session 22, “Proof by induction” of unit II of PEU 3300/PUU1140.

[Type text]

For your convenience we present here, the two versions of Principle of Mathematical induction as appeared in PEU 3300/PUU 1140.

Theorem (The Principle of Mathematical Induction)

Let $P(n)$ be any statement about n .

Assume that (i) $P(1)$ is true

and (ii) for each natural number n , if $P(n)$ is true then $P(n + 1)$ is true.

Then for each natural number n , $P(n)$ is true.

Theorem (Strong Form of Principle of Mathematical Induction)

Let $P(n)$ be a statement about n .

Assume that (i) $P(1)$ is true,

(ii) for each natural number n , if $P(1), P(2), \dots, P(n)$ are all true, then $P(n + 1)$ is true.

Then for each natural number n , $P(n)$ is true.

Though the two versions are equivalent, from the following activities and from the review questions appeared at the end of this session, you may realized the importance of both versions of the Principle of Mathematical Induction.

Activity 5



Let $s_0 = a$ and $s_n = 2s_{n-1} + b$ for $n \in \mathbb{N}$, where a and b are real constants.

(a) Calculate the first five terms of s_n .

(b) By observing these, can you guess a formula for s_n ? {depend on a, b and n }

(c) Use suitable version of Principle of Mathematical Induction to prove your guess.

Activity 6



Let $f_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right]$ for $n \in \mathbb{N}$. Using a suitable version of mathematical induction, prove that f_n is an integer for each $n \in \mathbb{N}$.

Review Questions

01. Prove that between any two real numbers there is a rational number.
02. Prove that between any two real numbers there is an irrational number.
03. Let m be an odd positive integer. Prove that $\sqrt{2m}$ is irrational.
05. Let $A = \{a + b : a, b \text{ are positive integers such that, } (a^2 + b^2) \text{ is a multiple of } 73\}$ and

Prove that the set A has a least element.

06. Let $B = \{b : b \text{ is a positive integer such that } (b^2 - 2) \text{ is a multiple of } 3\}$. Prove that the set A does not have a least element.
07. Prove that for each $n \in \mathbb{N}$, $1(1!) + 2(2!) + 3(3!) + \dots + n(n!) = (n+1)! - 1$.
08. Prove that for each $n \in \mathbb{N}$, $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$.
09. Prove that for each $n \in \mathbb{N}$, $\frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{n^2} \leq 2 - \frac{1}{n^2}$.
10. For $n \in \mathbb{N}$, prove (a) $\sum_{i=1}^n \frac{1}{\sqrt{i}} \geq \sqrt{n}$ and (b) $\sum_{i=1}^n \frac{1}{\sqrt{i}} \leq 2\sqrt{n} - 1$.
11. Prove that $8^{n+2} + 9^{2n+1}$ is divisible by 73 for $n \in \mathbb{N}$.
12. Prove that for each $n \geq 4$, $n! \geq 2^n$.